1. **What will be the output of this code?**

```python
j = (x for x in range(3))
for i in j:
    print(i**2, end=',')

for i in j:
    print(i, end=',')
```

A. 0,1,4,0,1,2,
B. 0,1,4,
C. 0,1,4,0,1,
D. No output is printed

What will be the output of the following Python code?

```python
lst = [1, 2, 3]
gen = (x * 2 for x in lst)
lst.append(4)
lst[0] = 10

print("List:", lst)
print("Generator output:", list(gen))
```

A. List: [10, 2, 3, 4]
Generator output: [2, 4, 6, 8]

B. List: [10, 2, 3, 4]
Generator output: [20, 4, 6, 8]

C. List: [10, 2, 3, 4]
Generator output: [2, 4, 6]

D. List: [1, 2, 3]
Generator output: [2, 4, 6]

2. What will be the output of the following code?

```python
[ ]: d = {0: [], 1: []}
     ref = d[0]
     for i in range(3):
         ref.append(i)
         d[i % 2].append(i * 10)

     print(d)
```

A. {0: [0, 1, 2], 1: [10, 20]}
B. {0: [0, 0, 1, 2, 20], 1: [10]}
C. {0: [0, 10, 1, 20, 2, 0], 1: [20]}
D. {0: [0, 1, 2, 10, 20], 1: [1, 2]}

3. What will be the output of the following recursive Python function?

```
def mystery(n):
    if n == 0:
        return 0
    result = mystery(n - 1)
    print(n, end=',')
    return result + n

print("Return Value:", mystery(3))
```

A. 1,2,3,
Return Value: 6
B 3,2,1,
Return Value: 6
B. 3,2,1,
Return Value: 0
C. 1,2,3,
Return Value: 0

4. What will be the output of the following code?

```
class Test:
    x = 0
    def __init__(self):
        self.x += 1
a = Test()
b = Test()
print(a.x, b.x, Test.x)
```

A. 1 1 0
B. 0 0 0
C. 1 1 1
D. 1 1 2

5. What will happen when the following code is executed?

```
x = 10
def func():
    print(x)
    x = 5
func()
```

A. 10
B. 5
C. 10 followed by 5
D. Error

6. What will be the output of the following code?

```
import numpy as np
a = np.array([[1], [2], [3]])
b = np.array([10, 20, 30])
print(a + b)
```

A. [[11]
[22]
[33]]

B. [[11 21 31]
[12 22 32]
[13 23 33]]

C [[11 21 31]]
[[12 22 32]]
[[13 23 33]]

D Error due to shape mismatch

7. What is the rank of the following matrix?

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 4 & 5 & 6 \end{bmatrix}$$

A. 3
B. 2
C. 1
D. 0

8. Given the DataFrame below, which of the following code snippets correctly selects columns 'A', 'B', and 'C'?

```
import pandas as pd
df = pd.DataFrame({
    'A': [1, 2, 3],
    'B': [4, 5, 6],
    'C': [7, 8, 9]
})
```

A. df['A', 'B', 'C']
B. df[['A', 'B', 'C']]
C. df[[['A', 'B', 'C']]]
D. df.loc[:, ['A', 'B', 'C']]

9. What happens when you run this code?

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3], [4, 5, 6])
plt.title("My Plot")
plt.show()
plt.title("New Title")
plt.show()
```

A. Two plots displayed, the second with the new title
B. One plot displayed twice with the same title
C. Only the first plot is displayed; the second plot shows no title
D. Error because title cannot be changed after show()

10. What will be the output of this code?

```
import seaborn as sns
import pandas as pd
df = pd.DataFrame({
    'x': [1, 2, 2, 3],
    'y': [4, 5, 6, 7]
})
sns.barplot(x='x', y='y', data=df);
```

A. A bar plot with bars for each unique 'x', showing mean of 'y' values
B. A bar plot with bars for each row in df
C. Error because 'x' has duplicate values
D. A scatter plot of all points

11. What will be the output of the following expression if df is a DataFrame containing an 'age' column?

```
df['age'] >= 30
```

A. Error
B. A single boolean value (True or False)
C. A Boolean Series indicating which rows have age greater than or equal to 30
D. A DataFrame containing only the rows where age is greater than or equal to 30

12. What does the following command return in pandas?

```
df.describe(include='all')
```

A. Descriptive statistics for only numeric columns
B. Descriptive statistics for only categorical columns
C. Descriptive statistics for all columns, including numeric, categorical, and object types
D. Error, because include='all' is not valid

13. What is the key difference between percentage and percentile?

A. Percentage is a measure of position in a dataset, while percentile is a part of a whole.
B. Percentage tells how much of a whole is achieved; percentile tells what portion of data scores fall below a specific value.
C. Percentile is always higher than percentage.
D. Percentage and percentile are the same and can be used interchangeably.

14. You are working with a dataset where the target variable is categorical (e.g., "Yes" or "No"), and you want to evaluate the relationship between each numerical feature and this target.

Which statistical test is most appropriate for feature selection in this case?

A. Chi-square Test
B. ANOVA (Analysis of Variance)
C. Pearson Correlation
D. Mutual Information for Discrete Data

15. You are training a linear regression model with many correlated features (multicollinearity). Which regularization technique is most suitable to both reduce overfitting and perform feature selection by shrinking some coefficients exactly to zero?

A. Ridge Regression (L2 Regularization)
B. Lasso Regression (L1 Regularization)
C. Elastic Net (Combination of L1 and L2)
D. Principal Component Regression

16. You trained a classification model that outputs probabilities. You want to find the best threshold to convert probabilities into class labels, optimizing for balanced precision and recall. Which method is the most appropriate to select this threshold?

A. Use the default threshold of 0.5
B. Plot and analyze the ROC curve
C. Plot and analyze the Precision-Recall curve
D. Compute accuracy for multiple thresholds and pick the highest

17. Is multicollinearity always bad in machine learning, and how can it be detected?
A. Multicollinearity is not always bad; it mainly affects interpretability rather than model performance in predictive tasks.
B. High multicollinearity inflates variance of coefficients, making them unreliable for interpretation.
C. It can be detected using Variance Inflation Factor (VIF), where a VIF value above 10 often indicates serious multicollinearity.
D. The p-values of predictors can also reveal multicollinearity if they are low but the R^2 value is high.

18. You are training a deep neural network using the sigmoid activation function in all hidden layers. Which issue is most likely to affect your model's training?

A. The network will always converge faster due to sigmoid's smoothness.
B. The vanishing gradient problem may occur because sigmoid outputs saturate at extreme values.
C. The exploding gradient problem will happen because sigmoid outputs are unbounded.
D. Sigmoid activation will cause the network to ignore bias terms.

19. Which of the following is NOT a benefit of using batch normalization in deep neural networks?

A. It helps reduce internal covariate shift by normalizing layer inputs.
B. It allows for higher learning rates and faster training.
C. It eliminates the need for activation functions in the network.
D. It acts as a regularizer, sometimes reducing the need for dropout.

20. Which of the following statements about gradient descent is FALSE?

A. Stochastic Gradient Descent (SGD) updates the model parameters using only one training example at a time.
B. Batch Gradient Descent computes the gradient using the entire training dataset before updating parameters.
C. Mini-batch Gradient Descent is a compromise between SGD and Batch Gradient Descent.
D. Increasing the learning rate always guarantees faster convergence without any risks.

21. You apply a 3×3 convolutional filter with stride 2 and no padding to a 7×7 input image. What will be the size of the output feature map?

A. 3×3
B. 2×2
C. 4×4
D. 5×5

22. Why are convolutional layers in CNNs preferred over fully connected layers for image data?

A. Because convolutional layers reduce the number of parameters by sharing weights and exploit spatial locality.
B. Because convolutional layers ignore spatial information, making training faster.
C. Because fully connected layers cannot learn nonlinear patterns.
D. Because convolutional layers always produce smaller output sizes than input sizes.

24. During GAN training, if the discriminator becomes too strong too quickly, what is the most likely consequence for the generator?

A. To generate realistic fake data from random noise.
B. To distinguish between real and fake data samples.
C. To minimize the reconstruction error of generated data.
D. To optimize the generator's architecture.

25. Which of the following is a major disadvantage of traditional Recurrent Neural Networks (RNNs)?

A. They cannot process sequential data of variable length.
B. They suffer from vanishing and exploding gradient problems, making it hard to learn long-term dependencies.
C. They require more memory than feedforward networks for the same input size.
D. They are unable to handle any type of non-linear data.

26. If an LSTM's forget gate is always set to 1 and the input gate is always set to 0, what happens over time?
A. The memory cell keeps accumulating new information.
B. The memory cell resets at each timestep.
C. The memory cell retains its initial state indefinitely.
D. The memory cell outputs random noise.

27. Which of the following best describes a key difference between LSTM and GRU architectures?

A. GRU uses three gates while LSTM uses two gates.
B. GRU merges the forget and input gates into a single update gate.
C. GRU always uses more parameters than LSTM.
D. GRU can't handle long-term dependencies like LSTM.

28. Given the scaled dot-product self-attention formula used in Transformers:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V$$

A. (Q) and (K) are used to compute token embeddings; (V) stores token positions; (d) is the number of tokens in the input.
B. (Q) (Query) is compared with (K) (Key) to calculate attention scores; (V) (Value) holds the actual information to attend over; (d) is the dimensionality of Q, K, and V vectors.
C. (Q), (K), and (V) are identical vectors in all layers; (d) is the total number of layers.
D. (QK^\top) gives the final output; softmax is used to train the model; (V) adjusts model weights; (d) is dropout rate.

29. Which of the following best describes the advantage of using Multi-Head Attention in Transformer models?
A. It allows the model to attend to multiple positions with different learned projections, capturing diverse semantic relationships.
B. It reduces the computational cost of the attention mechanism by splitting into parallel heads.
C. It eliminates the need for positional encoding by encoding time information within each head.
D. It ensures that queries and keys are always orthogonal to improve stability.

30. In the original Transformer architecture (Vaswani et al., 2017), positional encoding is added to input embeddings to inject sequence order. Which of the following statements best explains why sinusoidal functions are used for positional encoding?

A. Sinusoidal functions reduce overfitting by regularizing position vectors across layers.
B. The periodic nature of sine and cosine functions allows the model to extrapolate to sequence lengths not seen during training.
C. Sine and cosine are used because they are orthogonal and hence improve model convergence.
D. Sinusoidal encoding helps skip-layer connections avoid gradient vanishing in deeper networks.